第六章　振动

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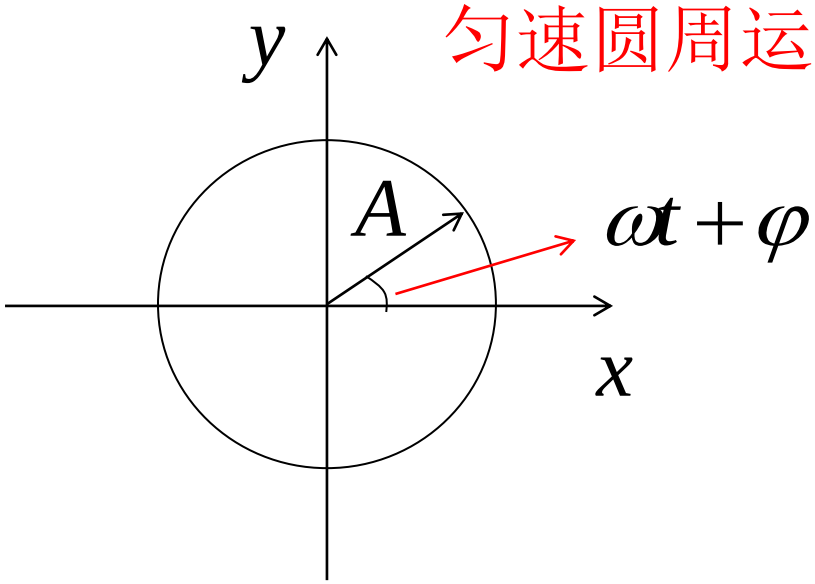
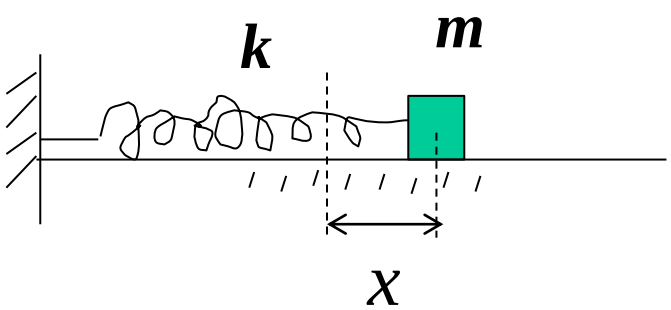
§6.1 简谐振动 （The simple harmonic oscillation）

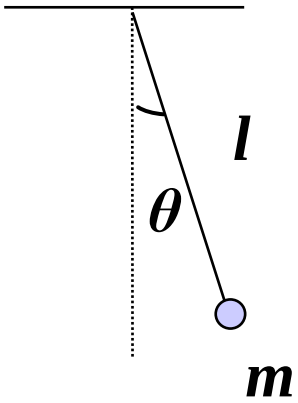
匀速圆周运动



初相

$$x = A\cos(\underbrace{\omega t + \varphi}_{\text{相}})$$





$\theta \longrightarrow x/l$

速度 $\upsilon = \dot{x} = -\omega A\sin(\omega t + \varphi)$

加速度 $a = \ddot{x} = -\omega^2 A\cos(\omega t + \varphi) = -\omega^2 x$

$$\boxed{\ddot{x} + \omega^2 x = 0}$$

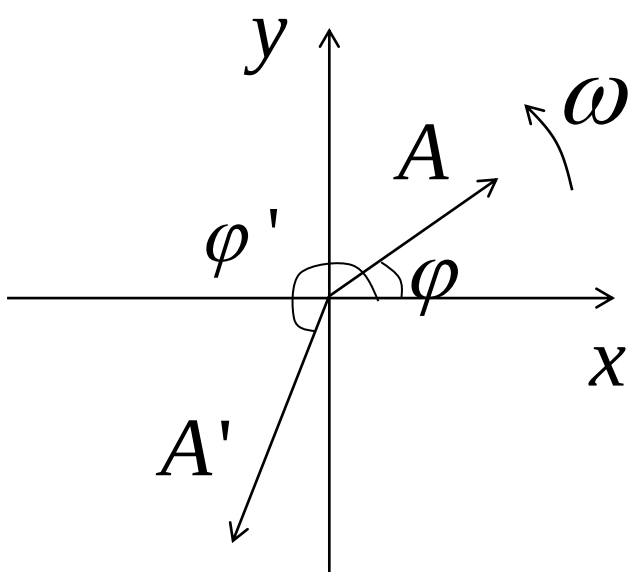
简谐振动方程

方程解 $x = A\cos(\omega t + \varphi)$

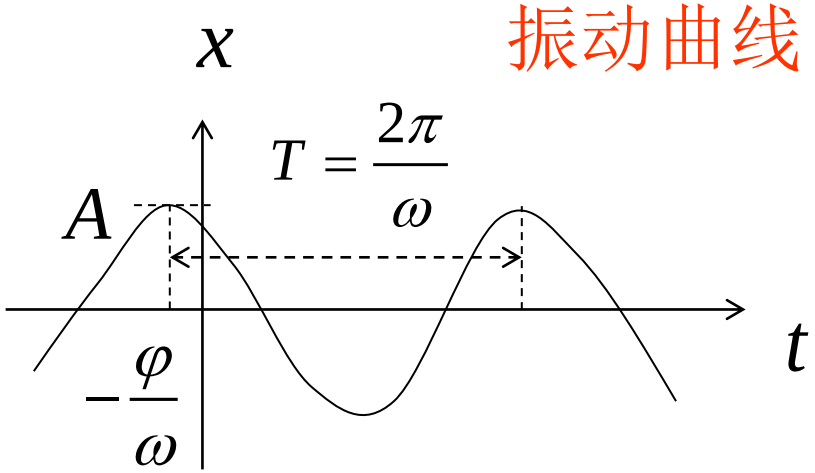
A, ω, φ 确定则简谐运动确定

$$x = A\cos(\omega t + \varphi)$$

相量图法



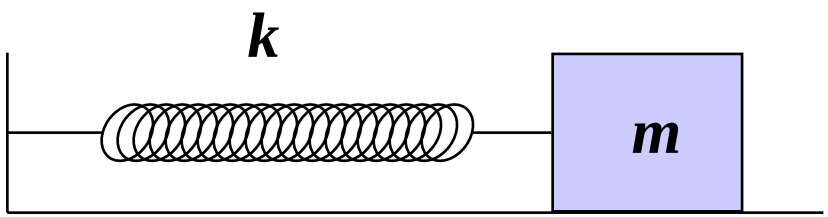
振动曲线



频率相同的简谐运动都可以在这里用几何图表示

§ 6.2 简谐振动方程

(i)

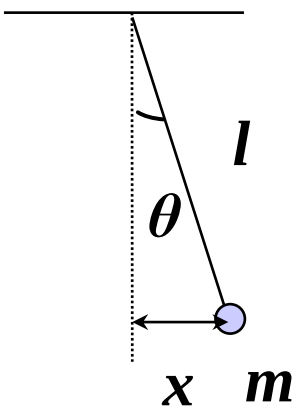


$$F = -kx = m\ddot{x}$$

$$\ddot{x} + \omega^2 x = 0$$

$$\omega = \sqrt{\frac{k}{m}}$$

(ii)



$$F = -mg\sin\theta$$

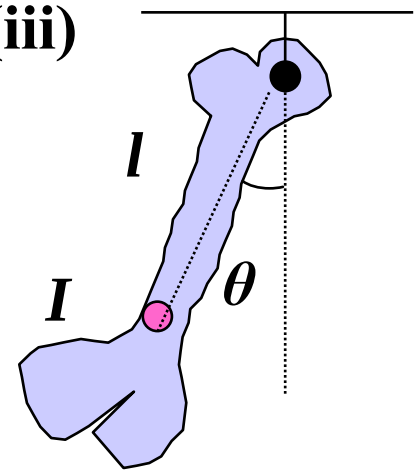
$$= -mg\frac{x}{l} \approx m\ddot{x}$$

$$\frac{x}{l} << 1$$

$$\ddot{x} + \omega^2 x = 0$$

$$\omega = \sqrt{\frac{g}{l}}$$

(iii)



$$\tau = -mgl\sin\theta$$

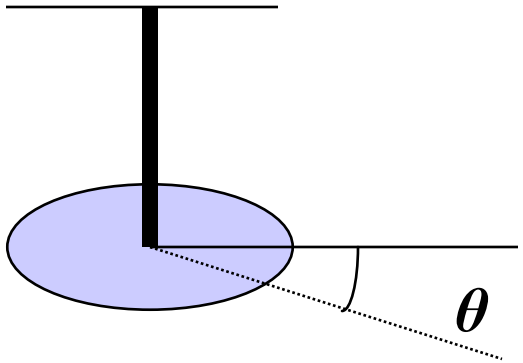
$$\approx -mgl\theta = I\ddot{\theta}$$

$$\theta << 1$$

$$\ddot{\theta} + \omega^2\theta = 0$$

$$\omega = \sqrt{\frac{mgl}{I}}$$

(iv)

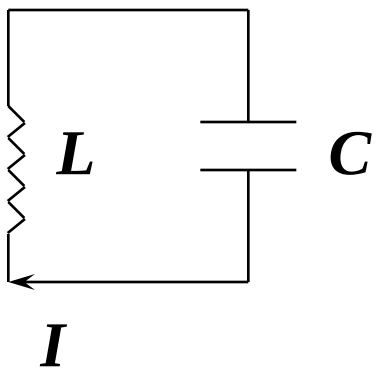


$$\tau = -\kappa\theta = I\ddot{\theta}$$

$$\ddot{\theta} + \omega^2\theta = 0$$

$$\omega = \sqrt{\frac{\kappa}{I}}$$

(v)



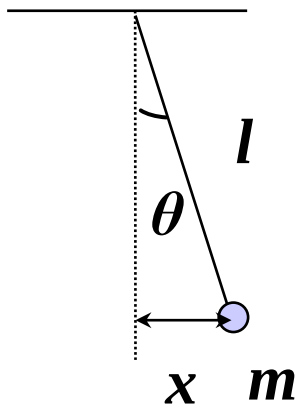
$$\mathcal{E} = -L\frac{dI}{dt} = -L\frac{d^2q}{dt^2} = \frac{q}{C}$$

$$\ddot{q} + \omega^2 q = 0 \qquad \omega = \sqrt{\frac{1}{LC}}$$

物理量 S $\qquad \ddot{S} + \omega^2 S = 0$ $\qquad$ 简谐振动

通解: $\qquad S = A\cos(\underbrace{\omega t + \varphi}_{\text{相位}})$ $\qquad$ 一般 A 大于零

例



1. 初始条件： $x|_{t=0} = x_m \qquad \dot{x}|_{t=0} = 0$

$$x = A\cos(\omega t + \varphi) \longrightarrow \varphi = 0$$

$$x = x_m \cos\omega t$$

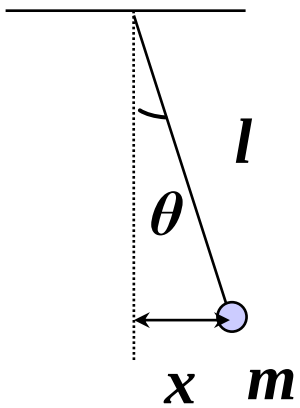
2. 初始条件： $x|_{t=0} = 0 \qquad \dot{x}|_{t=0} = \upsilon_0$

$$A\cos\varphi = 0 \qquad \upsilon_0 = -\omega A\sin\varphi$$

$$\dot{x} = -\omega A\sin(\omega t + \varphi) \qquad \varphi = -\frac{\pi}{2},\ \upsilon_0 = \omega A$$

$$x = \frac{\upsilon_0}{\omega}\cos(\omega t - \frac{\pi}{2}) = \frac{\upsilon_0}{\omega}\sin\omega t$$

§ 6.3 简谐振动能量



动能 $$E_K = \frac{1}{2}m\dot{x}^2 = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \varphi)$$

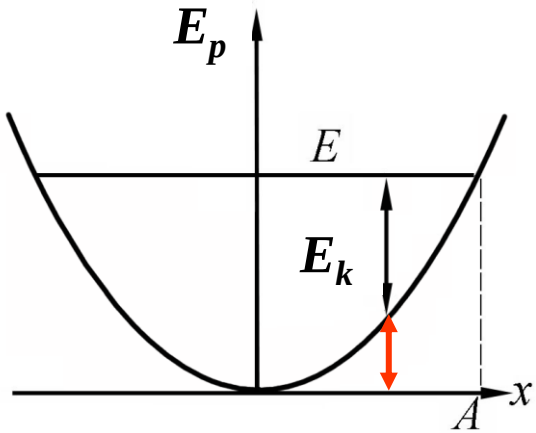
势能 $$E_P = mgl(1-\cos\theta) = 2mgl\sin^2\frac{\theta}{2}$$

$$E_P = \frac{1}{2}m\frac{g}{l}x^2 = \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2 A^2 \cos^2(\omega t + \varphi)$$

$$l^2 \sin^2\frac{\theta}{2} \approx \frac{x^2}{4}$$

$$E = E_K + E_P = \frac{1}{2}m\omega^2 A^2 = const.$$

$$E_P = \frac{1}{2}m\omega^2 x^2$$



时间平均 $$\left\langle \cos^2(\omega t+\varphi)\right\rangle = \frac{1}{T}\int_0^T \cos^2(\omega t+\varphi)dt = \frac{1}{2}$$

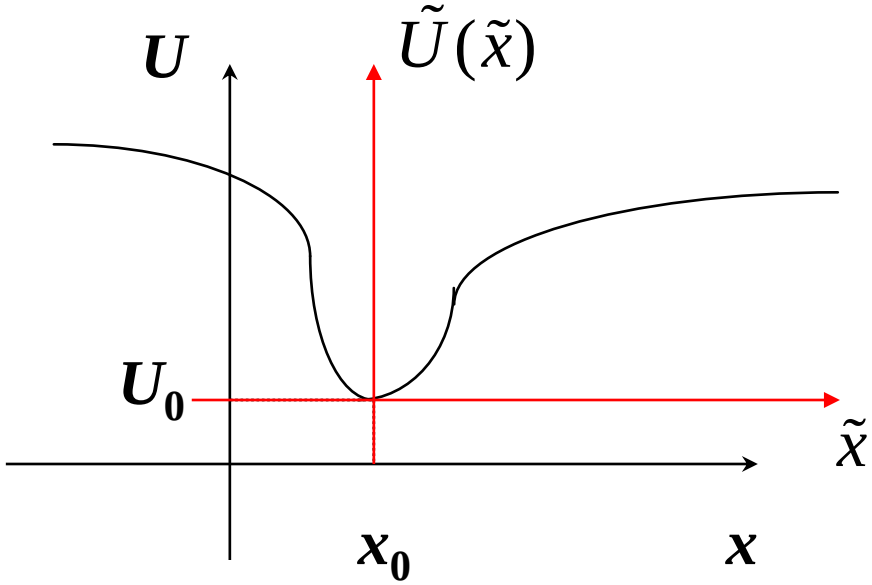
$$\left\langle E_P\right\rangle = \frac{1}{4}m\omega^2 A^2 = \left\langle E_K\right\rangle = \frac{1}{2}E$$

或 $$\begin{cases} E = \text{const}. \\ \left\langle E_{\mathrm{p}}\right\rangle = \left\langle E_{\mathrm{k}}\right\rangle = \dfrac{1}{4}m\omega^2 A^2 \propto A^2 \end{cases}$$

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2 \longrightarrow \ddot{x} + \omega^2 x = 0$$

$$E_p = \frac{1}{2}m\omega^2 x^2 \longrightarrow F_x = -\frac{\partial E_k}{\partial x} = -m\omega^2 x$$

势函数极小值附近的运动



$$|x - x_0| << 1$$

$$U(x) = U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2!}U''(x_0)(x - x_0)^2 + \cdots$$

U 极小 $\quad U'(x_0) = 0$

$U''(x_0) > 0$

$$U(x) \doteq U(x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$$

平移原点 $\quad x - x_0 = \tilde{x}$

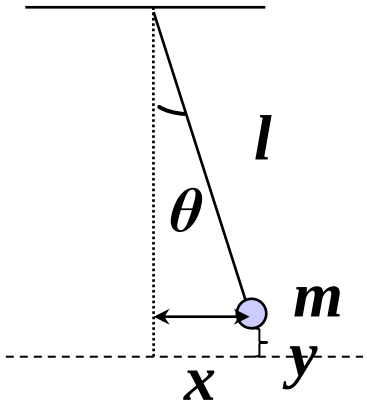
$$\tilde{U}(\tilde{x}) = U(x) - U(x_0)$$

$$\tilde{U}(\tilde{x}) \approx \frac{1}{2}\underline{\tilde{U}''(0)}\tilde{x}^2$$

$m\omega^2$ 或 k

微小振动一般都是简谐振动

例. 单摆



$U = 0$

$x = l\sin\theta$

$$U = mgy$$

$$= mgl(1-\cos\theta)$$

$$= mgl(1-\sqrt{1-\sin^2\theta})$$

$$= mgl(1-\sqrt{1-\frac{x^2}{l^2}})$$

$x << l$

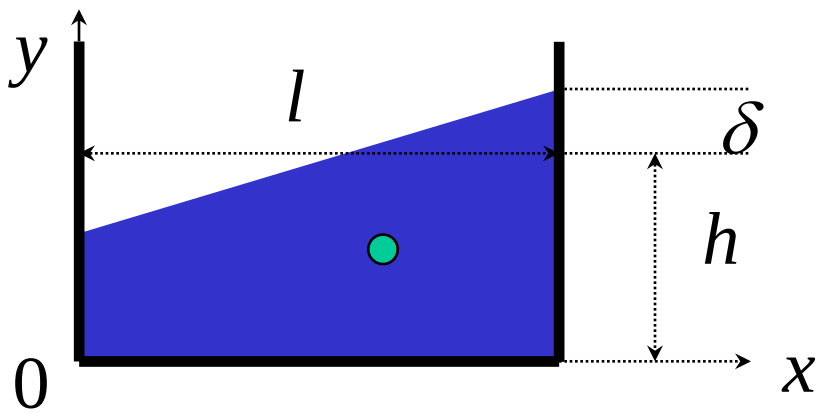
$$\doteq \frac{1}{2}m\frac{g}{l}x^2$$

$$U(x) \doteq \frac{1}{2}m\omega^2 x^2$$

$$\omega^2 = \frac{g}{l}$$

单摆微小振动角频率

例. 水盆中的晃动模式. 谁能够端一盆水而没有晃动呢？



晃动频率？

解：假设是微小振动

简谐振动?

势能表达式 $E_P = mg(y_c - \frac{h}{2})$

$$x_c = \frac{\delta \frac{2}{3} l + (h-\delta)\frac{1}{2} l}{h} = \frac{1}{2} l + \frac{1}{6} l \frac{\delta}{h}$$

$$y_c = \frac{[\frac{2}{3}\delta + (h-\delta)]\delta + \frac{1}{2}(h-\delta)^2}{h} = \frac{1}{2} h + \frac{1}{6}\frac{\delta^2}{h}$$

$$E_P = mg\frac{1}{6}\frac{\delta^2}{h} = 6mg\frac{h}{l^2}(x_c - \frac{1}{2}l)^2$$

$$= \frac{1}{2}k(x_c - \frac{l}{2})^2 = \frac{1}{2}m\omega^2(x_c - \frac{l}{2})^2$$

简谐振动 频率 $\omega^2 = 12g\frac{h}{l^2}$

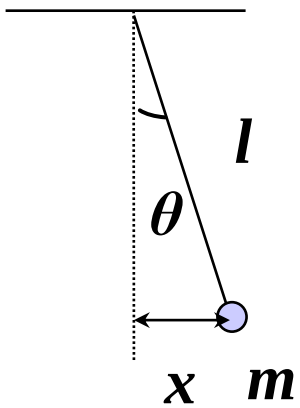
日内瓦湖的平均深度约150m，长度约60km，晃动周期？

$$T = \frac{2\pi}{\omega} = \frac{\pi l}{\sqrt{3gh}} \approx 47\,\text{min}.$$

实际观察到的周期约1小时

观察到的振幅大于5 feet~1.5m

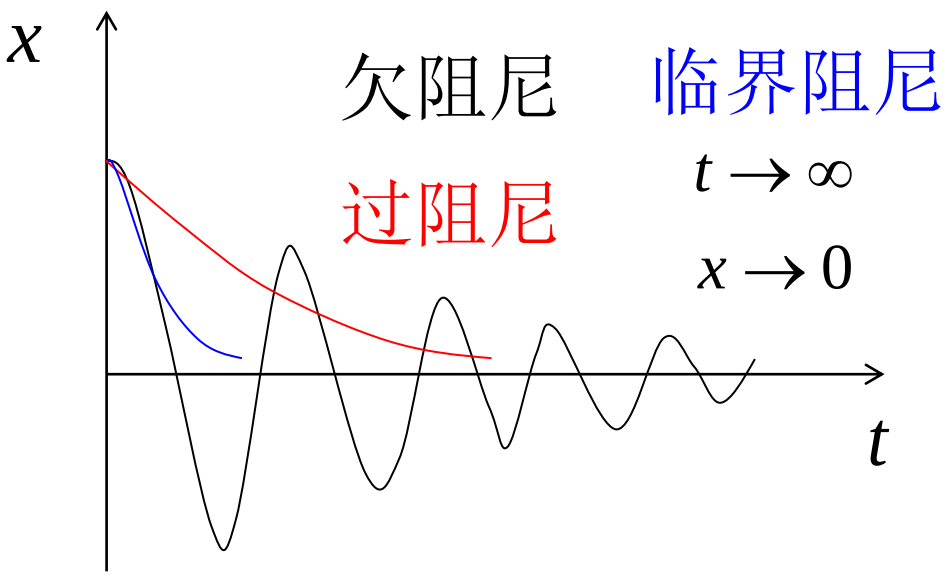
§ 6.4 阻尼振动



$$F = -mg\frac{x}{l} = -m\omega_0^2 x \qquad f \propto -\dot{x}$$

$$-m\omega_0^2 x - \gamma\dot{x} = m\ddot{x}$$

$$\boxed{\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0} \qquad \beta > 0$$



$$Ae^{-\beta t}\cos(\omega t + \varphi)$$

$$Ae^{-\beta t}(1 + \frac{t}{\tau})$$

$$e^{-\beta t}(Ae^{\omega t} + Be^{-\omega t})$$

$$\omega = \sqrt{\left|\beta^2 - \omega_0^2\right|}$$

品质因数 Q 值　　欠阻尼

振动能量　$W(t) \approx \frac{1}{2} m\omega^2 A^2 e^{-2\beta t}$

阻尼特征时间　$\tau = \frac{1}{2\beta}$

2π× 振动次数　$\boxed{Q = 2\pi \frac{\tau}{T} = \frac{\omega}{2\beta}}$

一个周期损耗能量　$W_R(t) \approx 2\beta T W(t)$

$$\boxed{\text{品质因数} = 2\pi \times \frac{\text{储存能量}}{\text{一个周期损耗能量}}}$$

损耗越小
Q值越大

§ 6.5 受迫振动

$$-m\omega_0^2 x - \gamma\dot{x} + F(t) = m\ddot{x}$$

周期性驱动力

$$F = H\cos\omega t$$

$$-m\omega_0^2 x - \gamma\dot{x} + H\cos\omega t = m\ddot{x}$$

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = h\cos\omega t$$

固有频率

非齐次方程

$x =$ 齐次解(暂态解)+特解(稳态解)

时间长了以后, 随驱动频率振动

非齐次方程稳态解

$$x = A\cos(\omega t + \varphi)$$

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = h\cos\omega t$$

$$(\omega_0^2 - \omega^2)A\cos(\omega t + \varphi) - 2\beta A\omega\sin(\omega t + \varphi) = h\cos\omega t$$

$$(\omega_0^2 - \omega^2)A\cos\varphi - 2\beta A\omega\sin\varphi = h$$

$$-(\omega_0^2 - \omega^2)A\sin\varphi - 2\beta A\omega\cos\varphi = 0$$

$$A=\frac{h}{(\omega_0^2-\omega^2)\cos\varphi-2\beta\omega\sin\varphi}$$

$$\tan\varphi=\frac{2\beta\omega}{\omega^2-\omega_0^2}$$

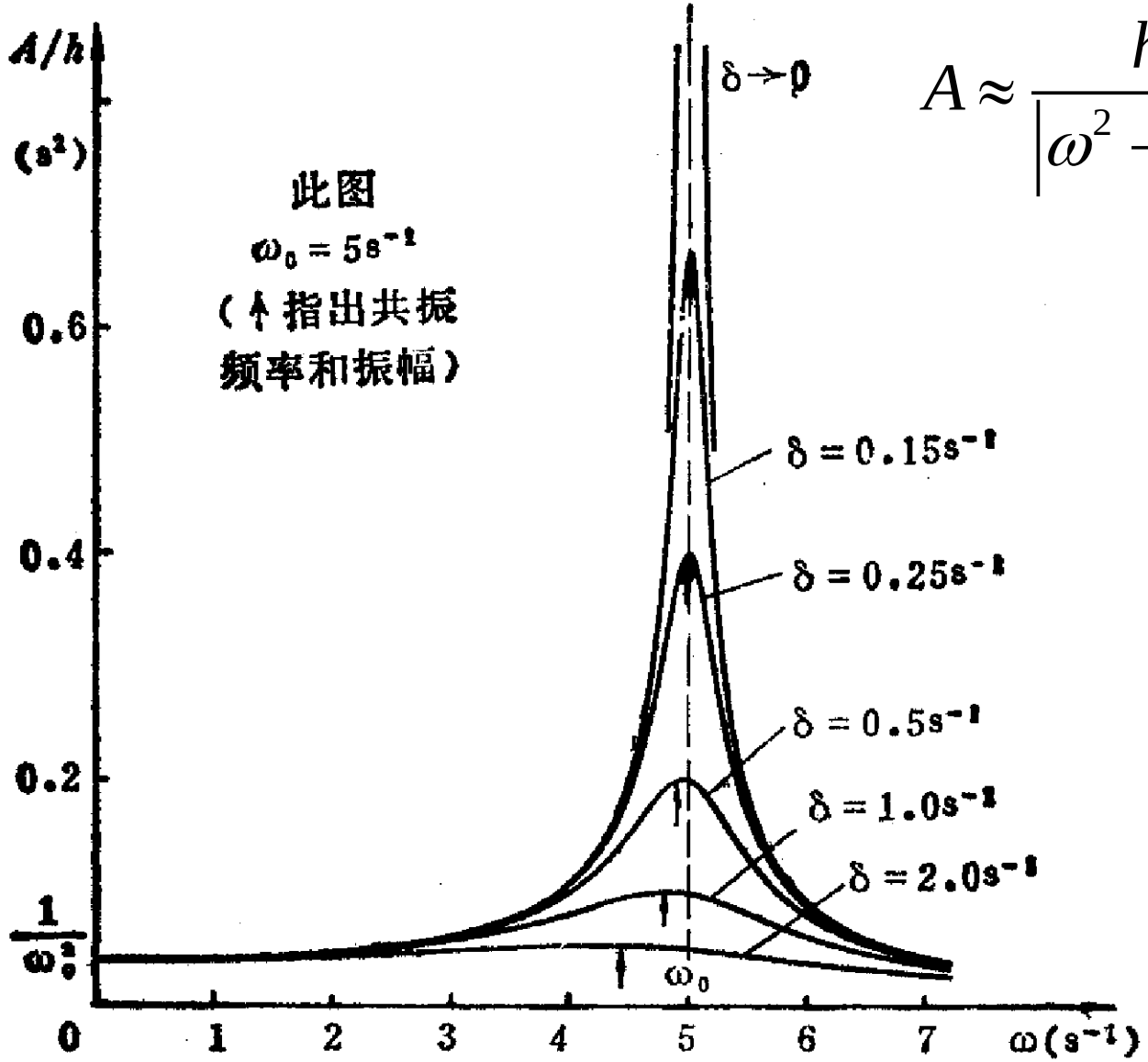
$$=\frac{h}{\sqrt{(\omega^2+2\beta^2-\omega_0^2)^2+4\beta^2(\omega_0^2-\beta^2)}}$$

弱阻尼情况 $\beta=\delta<<\omega_0$

$$A=\frac{h}{\sqrt{(\omega^2-\omega_\gamma^2)^2+4\gamma^2}}$$

共振

$\omega=\omega_\gamma\approx\omega_0$ 振幅最大 $A_{\max}=\frac{h}{2\gamma}$



$$A \approx \frac{h}{\left|\omega^2 - \omega_0^2\right|}$$

共振(曲线)
Resonance

不同阻尼的几条共振曲线

1940年华盛顿的塔科曼大桥建成

同年7月的一场大风引起桥的共振，使桥摧毁.

速度幅

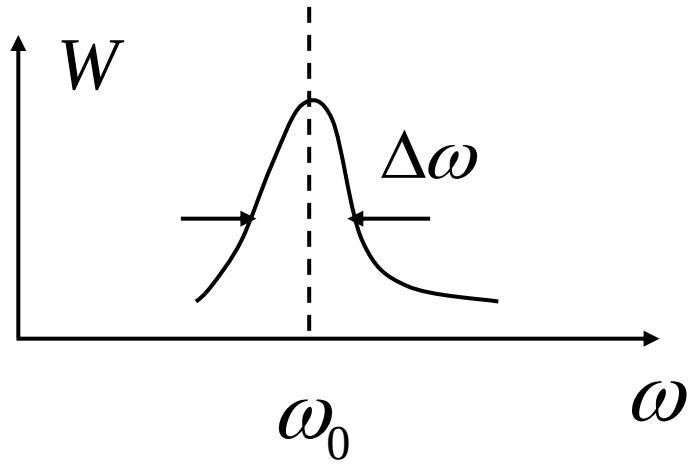
$$\upsilon=\omega A=\frac{\omega h}{(\omega_0^2-\omega^2)\cos\varphi-2\beta\omega\sin\varphi}$$

$$=\frac{h}{\sqrt{(\omega-\frac{\omega_0^2}{\omega})^2+4\beta^2}}$$

$\omega=\omega_0$ 速度共振

速度幅最大 $\upsilon_{\max}=\frac{h}{2\beta}$

品质因数与共振(谐振)曲线的锐度



半宽度 $\Delta\omega$

频率在谐振频率附近

$$\omega = \omega_0 \pm \frac{\Delta\omega}{2}$$

$$W \approx \frac{1}{2} m\omega^2 A^2$$

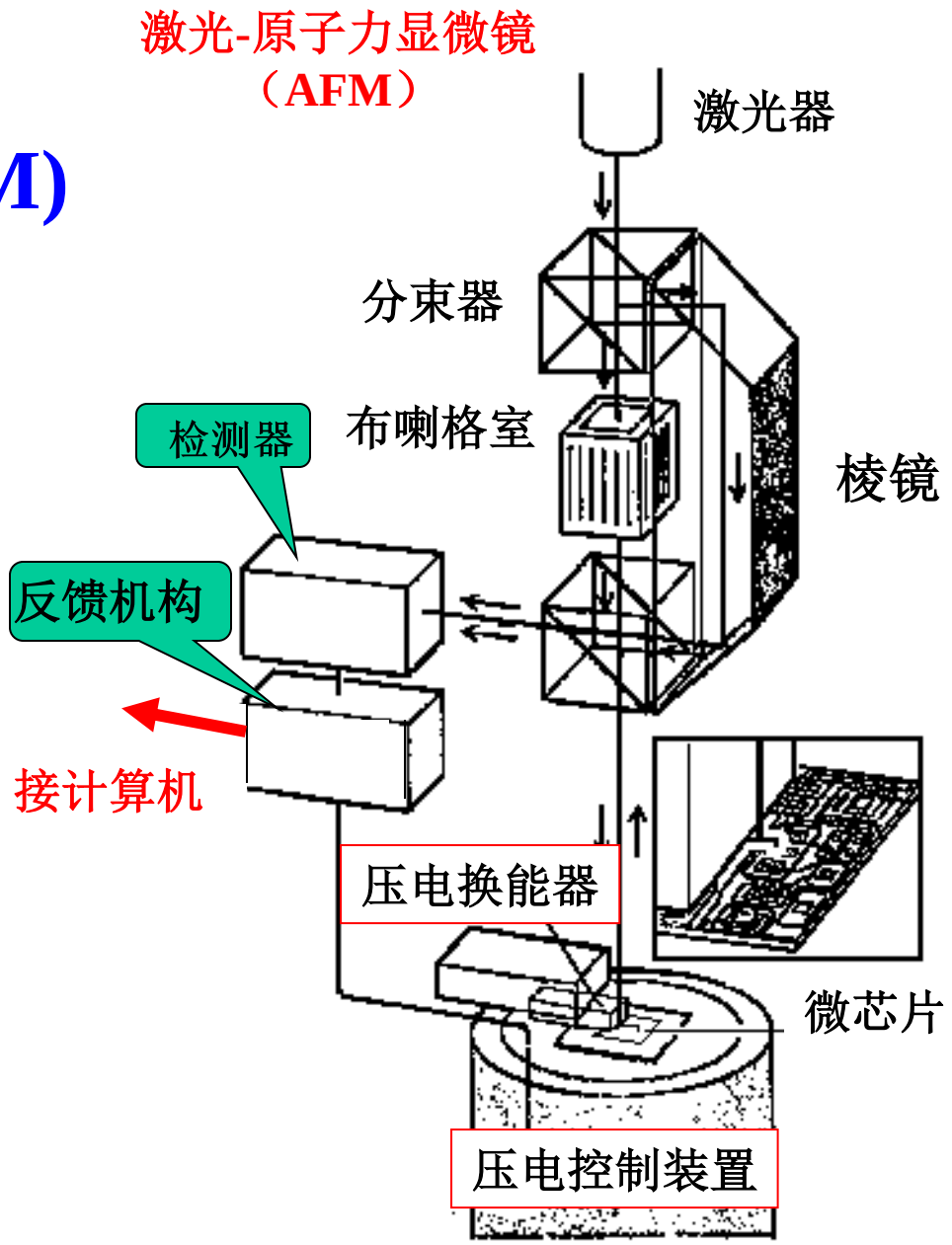
$$W \propto \frac{1}{(\omega - \frac{\omega_0^2}{\omega})^2 + 4\beta^2}$$

$$\Delta\omega \approx 2\beta$$ 降一半

$$\boxed{\frac{1}{Q} = \frac{\Delta\omega}{\omega_0}} \approx \frac{2\beta}{\omega_0} \qquad \omega \approx \omega_0$$

激光——
原子力显微镜(AFM)

一根钨探针或硅探针受原子力， 振动频率变化，由激光干涉探测到



试样通常是微电子器件。

挑战者号航天飞机失事

低温使O圈失效，火箭固有振动使O圈处产生缝隙，火焰泄露，高温烧穿燃料箱，发生爆炸。

$$\tau = -mgl\sin\theta$$

$$\approx -mgl\theta = I\ddot{\theta}$$

$$\theta << 1$$

$$\ddot{\theta} + \omega^2\theta = 0$$

$$\omega = \sqrt{\frac{mgl}{I}}$$

§ 6.6 同振动方向同频率简谐振动合成

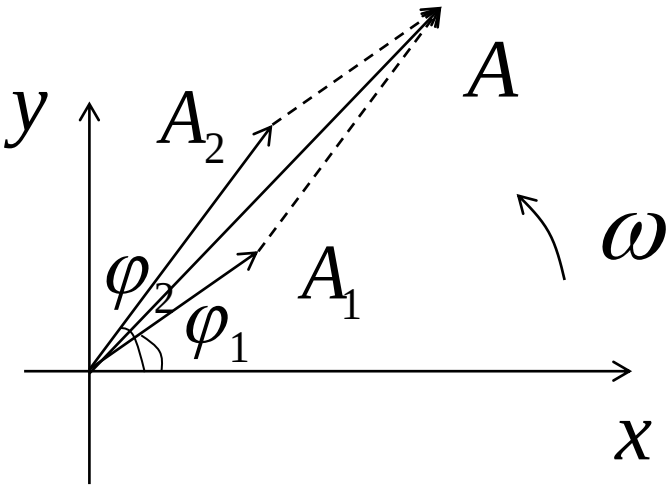
$$x_1 = A_1\cos(\omega t + \varphi_1)$$

$$x_2 = A_2\cos(\omega t + \varphi_2)$$

$$x = x_1 + x_2$$

$$= A\cos(\omega t + \varphi)$$

代数或几何法



$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_2 - \varphi_1)}$$

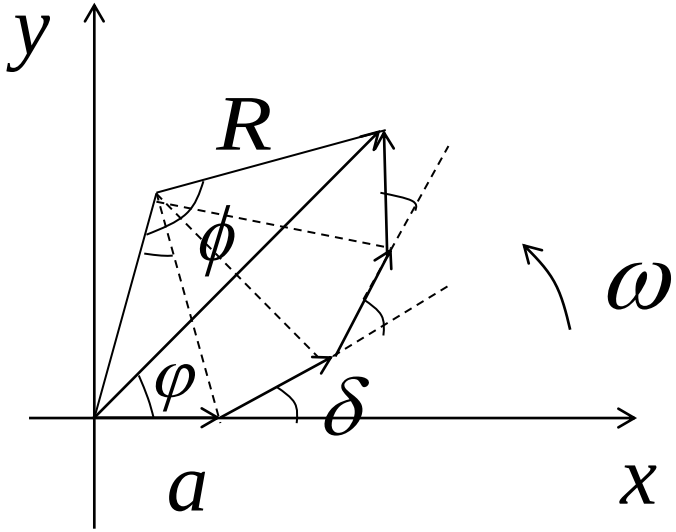
$$\varphi = \tan^{-1}\frac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2}$$

同振动方向同频率简谐振动合成简谐振动

应用: 双缝干涉

$$x=\sum_{i=1}^{n}x_i=\sum_{i=1}^{n}a\cos[\omega t+(i-1)\delta]$$

应用背景：光栅



等 N 边型一部分

$$\phi=n\delta$$

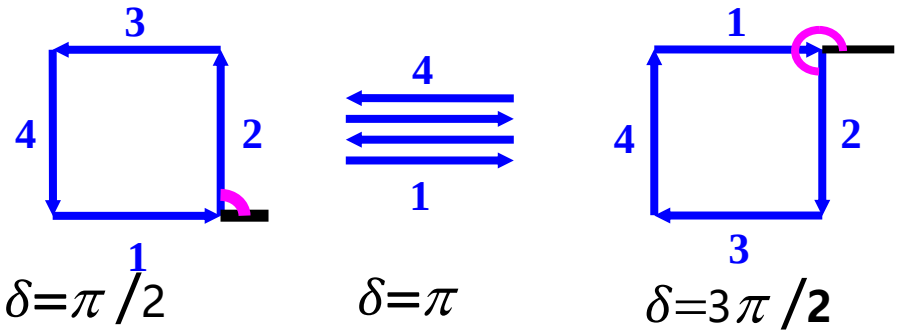
$$\varphi=\frac{(n-1)\delta}{2}$$

$$R\sin\frac{\delta}{2}=\frac{a}{2}\qquad R\sin\frac{\phi}{2}=\frac{A}{2}\qquad A=a\frac{\sin\frac{n\delta}{2}}{\sin\frac{\delta}{2}}$$

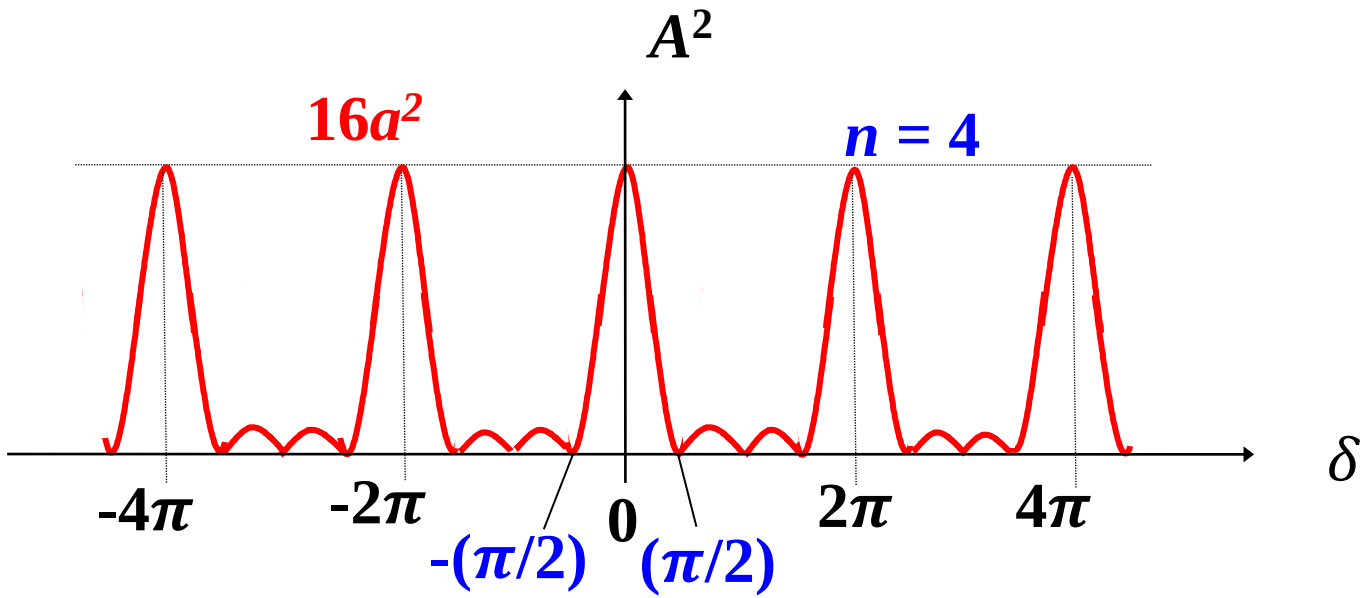
$$x=\sum_{i=1}^{n}x_i=A\cos(\omega t+\varphi)$$

$A=0$?

$n = 4$



$$A^2 = a^2 \frac{\sin^2 \frac{n\delta}{2}}{\sin^2 \frac{\delta}{2}}$$



§ 6.7 同振动方向不同频率简谐振动合成

$$x_1 = A_1 \cos(\omega_1 t + \varphi_1)$$

$$x_2 = A_2 \cos(\omega_2 t + \varphi_2)$$

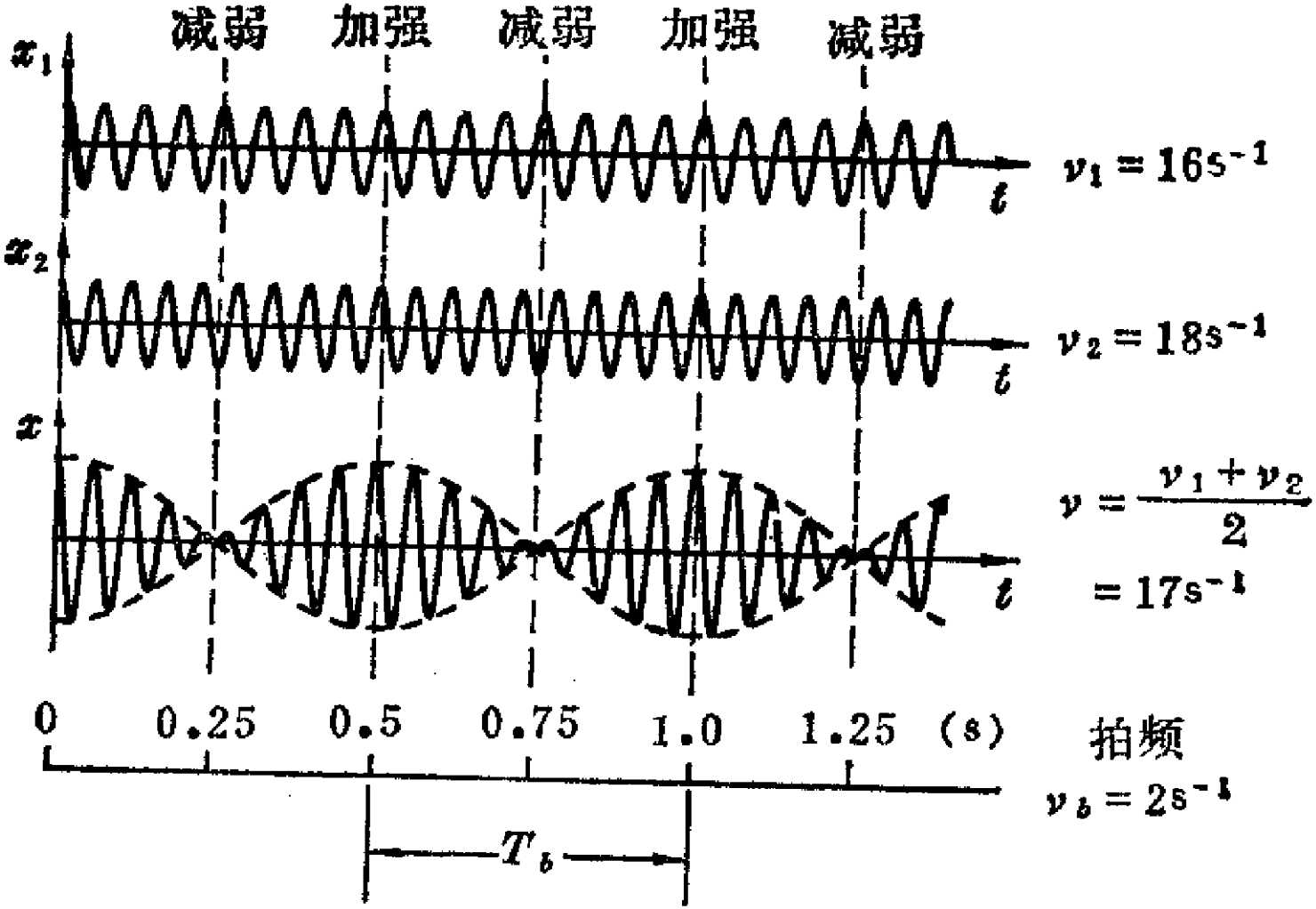
$$x = x_1 + x_2$$

$$= 2A\cos\left(\frac{\omega_2 - \omega_1}{2} t + \frac{\varphi_2 - \varphi_1}{2}\right)\cos\left(\frac{\omega_2 + \omega_1}{2} t + \frac{\varphi_2 + \varphi_1}{2}\right)$$

不再是简谐振动

频率接近 $\omega_2 - \omega_1 << \omega_2 + \omega_1$

拍 拍频 $\omega = 2 \times \frac{\omega_2 - \omega_1}{2} = \omega_2 - \omega_1$



二频率相近的谐振动合成的一例

§ 6.8 *频谱分析

$$a_0 = \frac{2}{T}\int_0^T f(t)dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\frac{2\pi}{T}tdt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\frac{2\pi}{T}tdt$$

任何周期振动

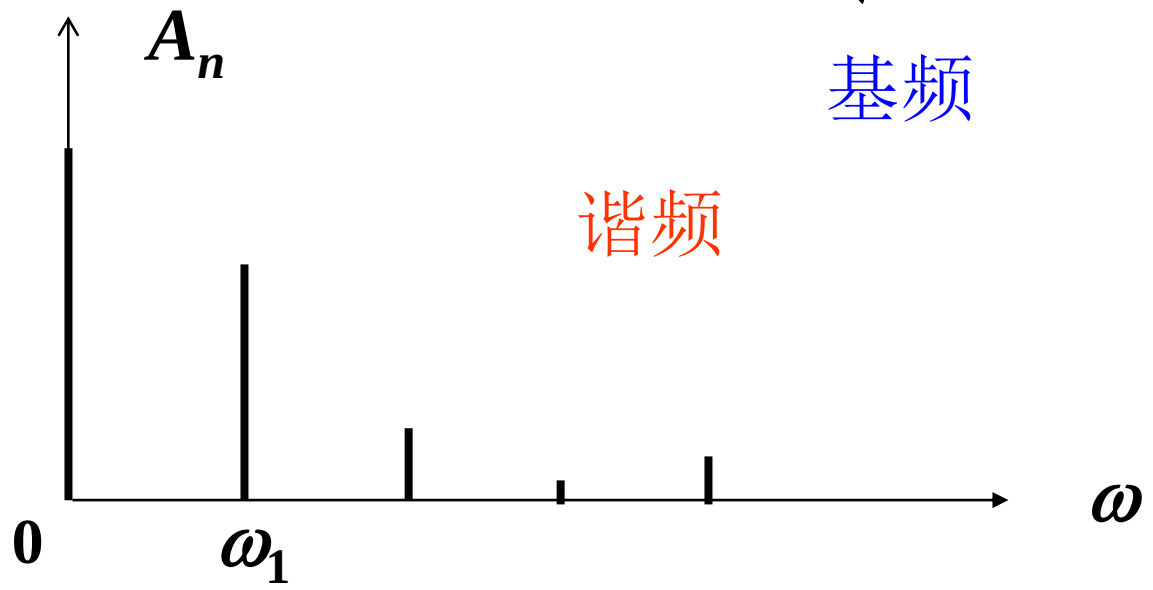
$$f(t) = \frac{a_0}{2} + \sum_{n=1}(a_n \cos\omega_n t + b_n \sin\omega_n t)$$

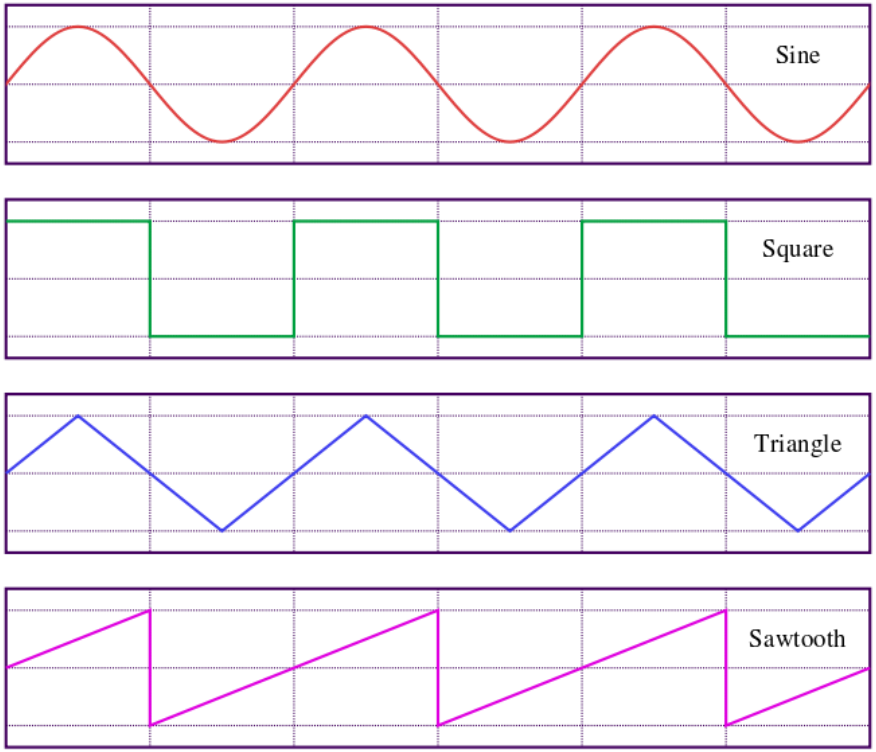
$$= \frac{a_0}{2} + \sum_{n=1} A_n \cos(\omega_n t + \phi_n) \qquad \omega_n = \frac{2\pi n}{T} = n\omega_1$$

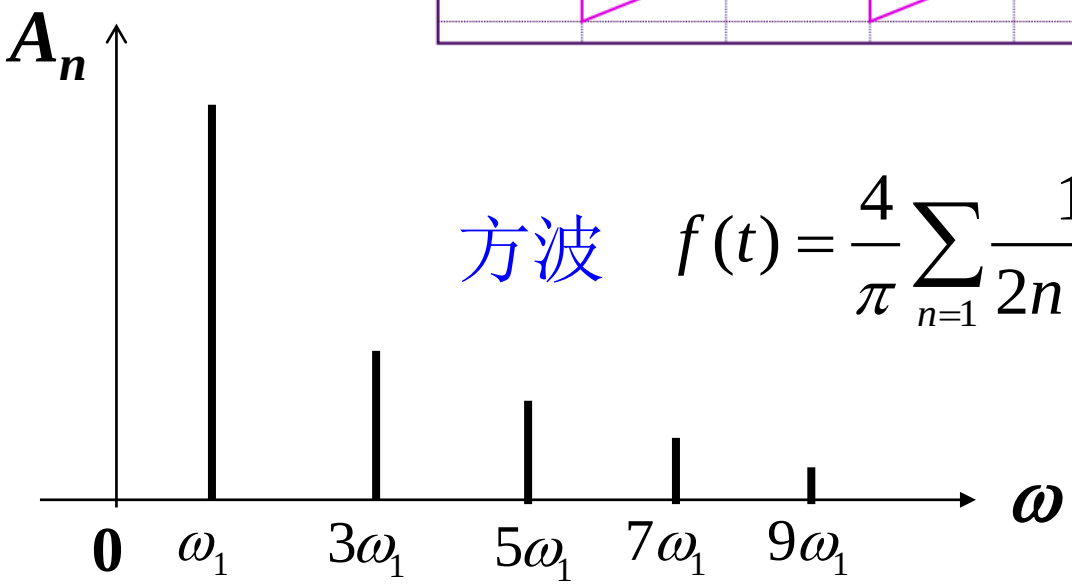
基频

简谐振动的叠加

频率是基频的整数倍数

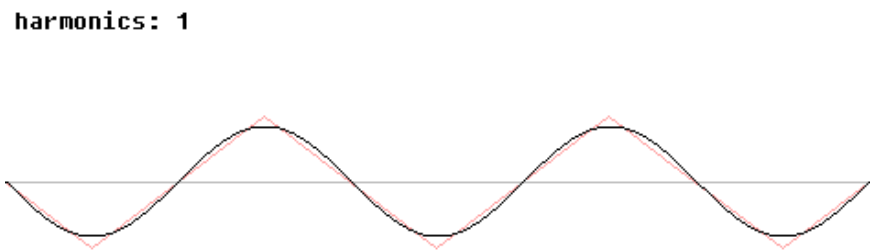




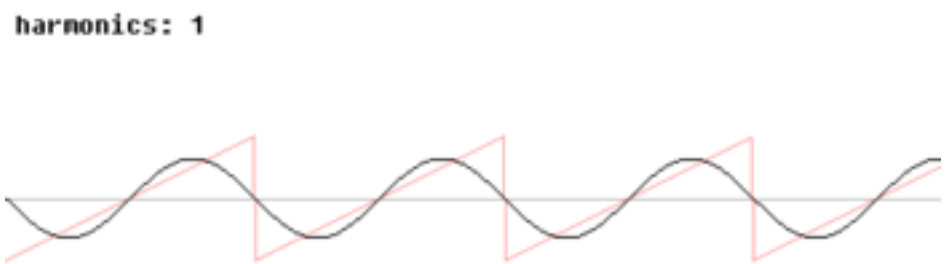


方波 $$f(t)=\frac{4}{\pi}\sum_{n=1}\frac{1}{2n-1}\sin[(2n-1)\omega t]$$

三角波 $$f(t)=\frac{8}{\pi^2}\sum_{n=1}\frac{(-1)^{n+1}}{(2n-1)^2}\sin[(2n-1)\omega t]$$



锯齿波 $$f(t)=\frac{2}{\pi}\sum_{n=1}\frac{(-1)^{n+1}}{n}\sin(n\omega t)$$

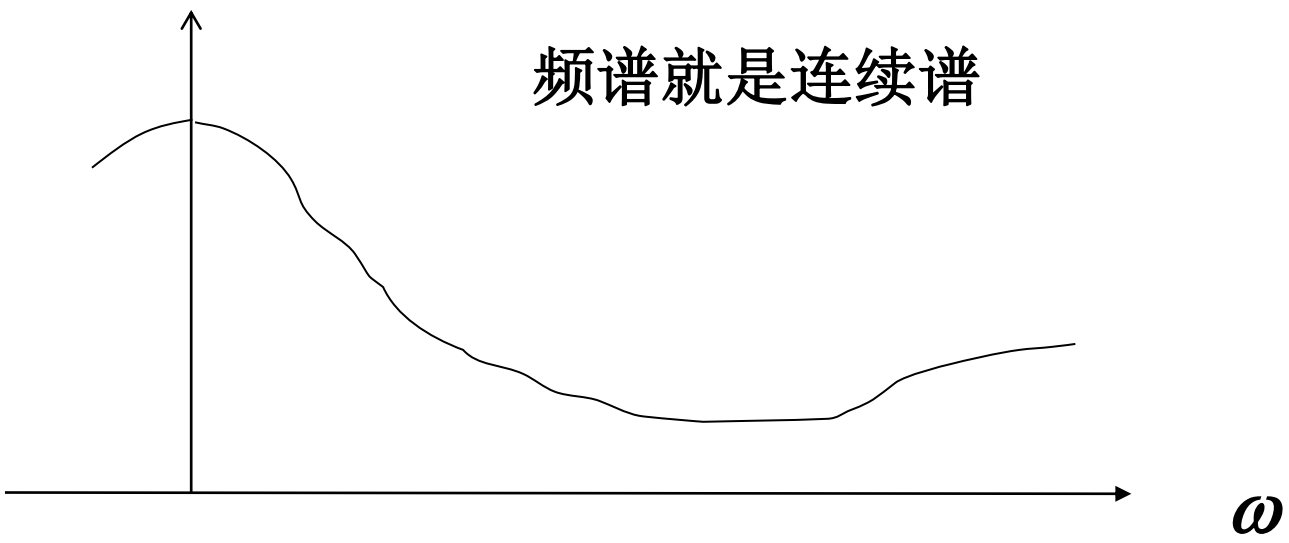


任何非周期运动，有傅里叶变换

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}F(\omega)e^{i\omega t}d\omega$$

$$F(\omega)=\int_{-\infty}^{+\infty}f(t)e^{-i\omega t}dt$$

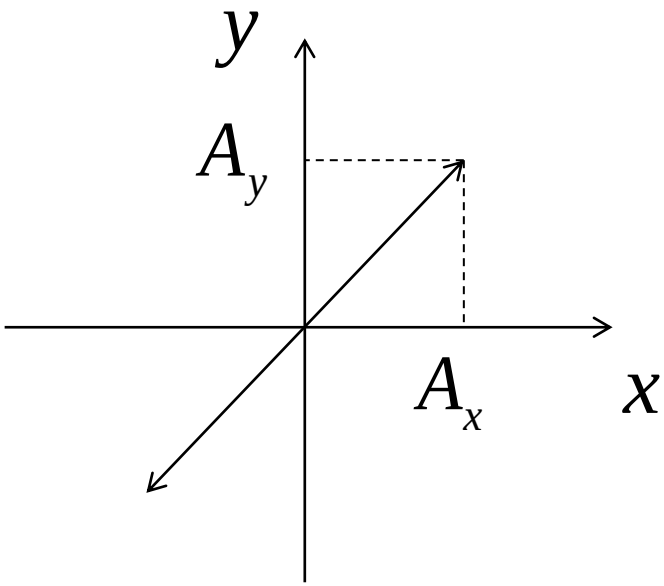
频谱就是连续谱



§ 6.9 振动方向相互垂直的简谐振动合成

$$x = A_x \cos(\omega_x t + \varphi_x)$$

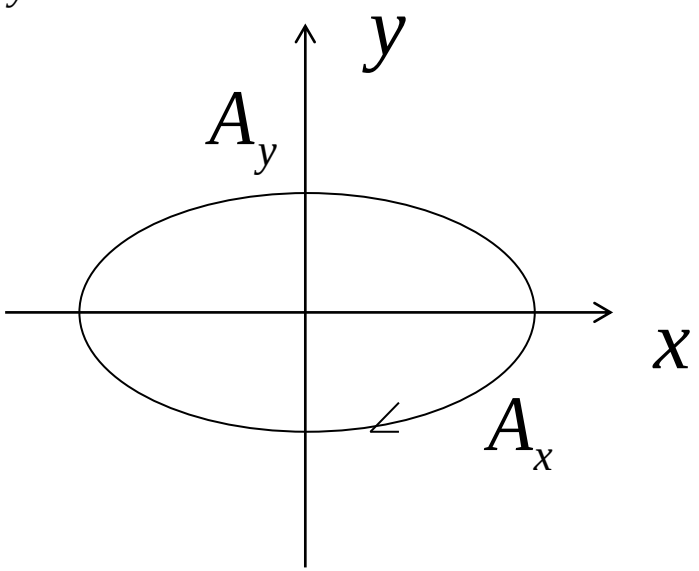
$$y = A_y \cos(\omega_y t + \varphi_y)$$



同频率

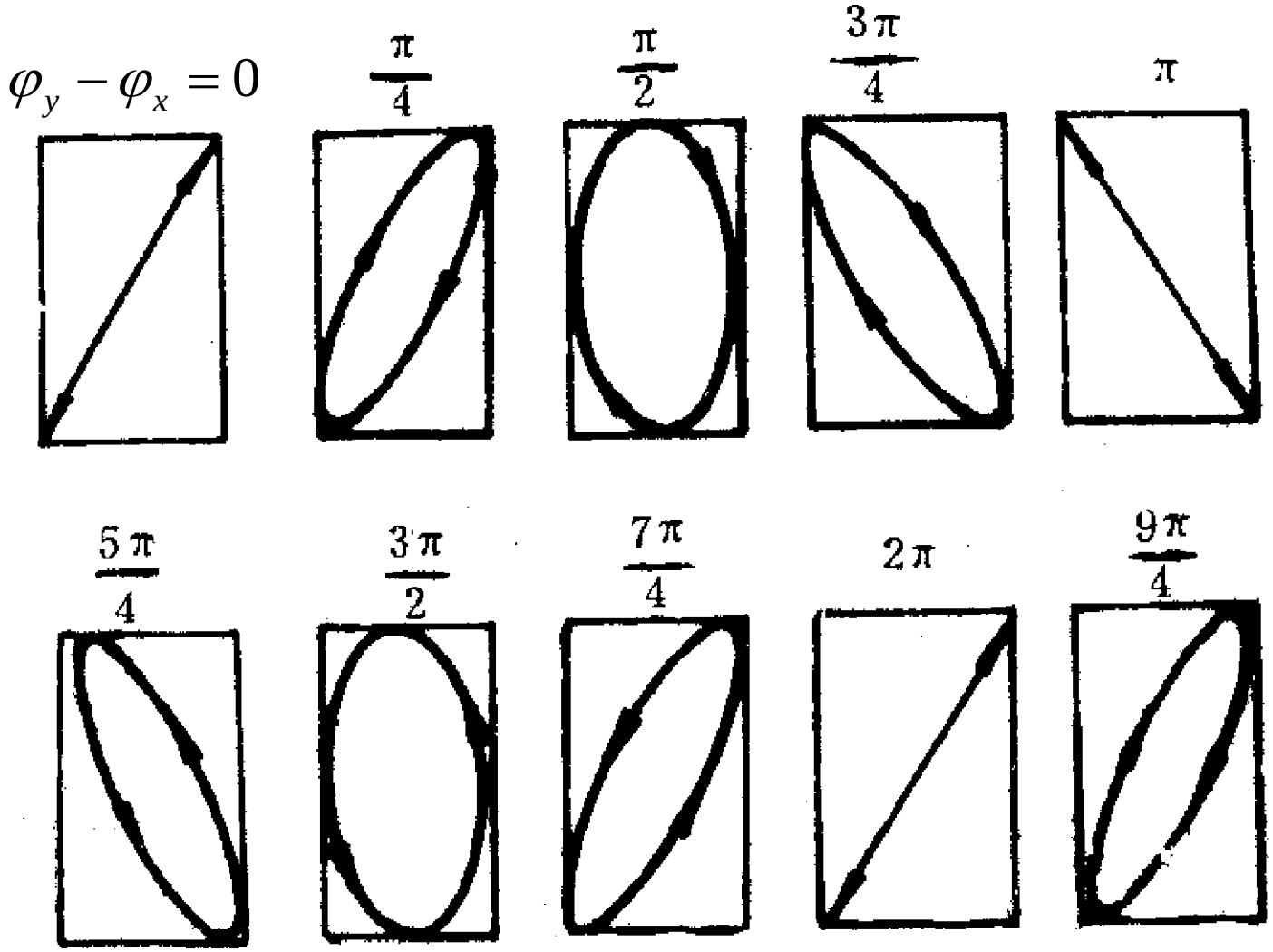
$$x = A_x \cos \omega t$$

$$y = A_y \cos \omega t$$



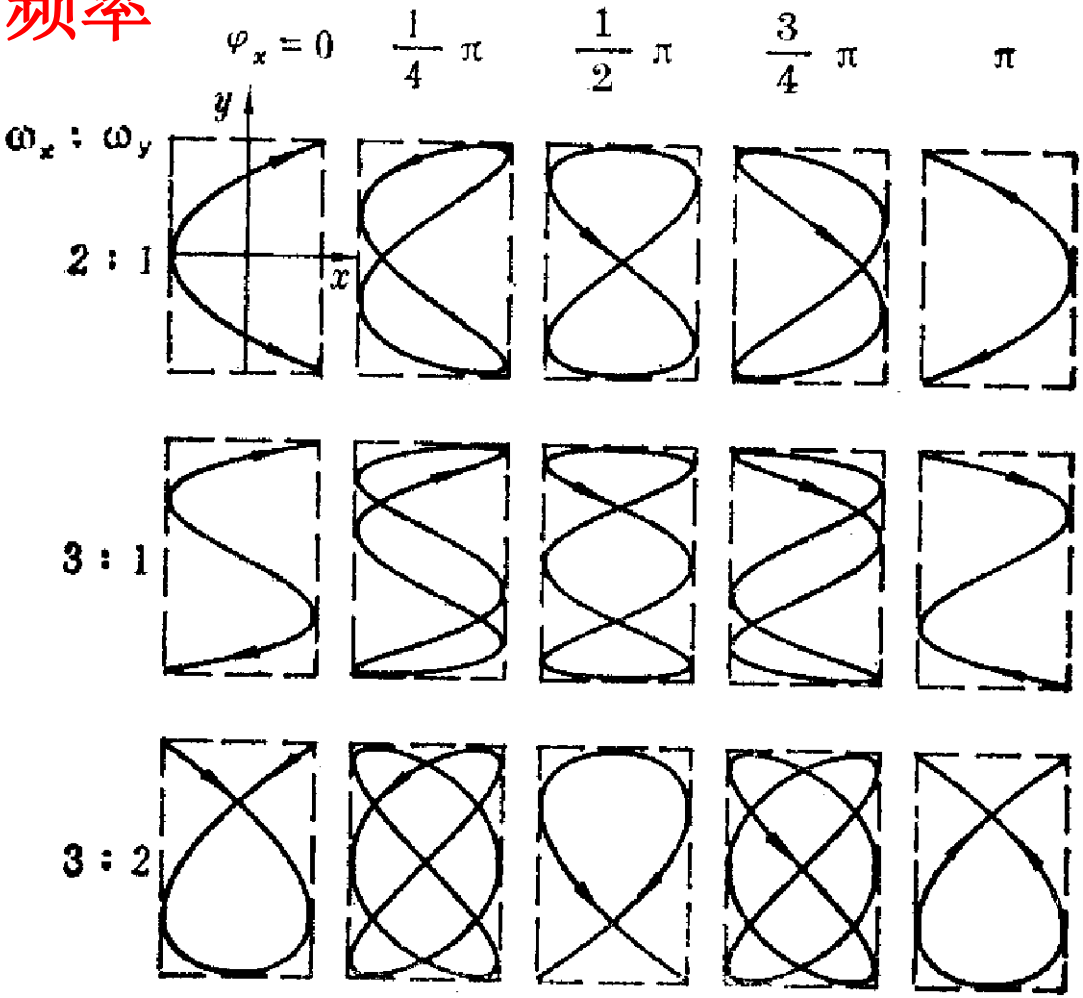
$$x = A_x \cos \omega t$$

$$y = A_y \cos(\omega t + \frac{\pi}{2})$$



两个垂直方向同频率简谐振动的合成

不同频率



利萨如图的一些例子

$x = A_x \cos(\omega_x t + \varphi_x)$

$y = A_y \cos \omega_y t$

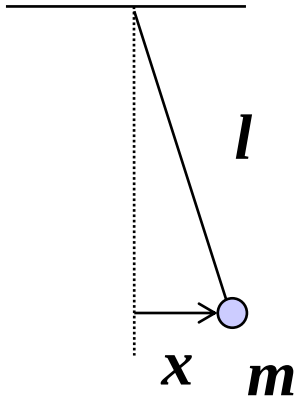
$A_x : A_y = 2:3$

若频率非整数比曲线不闭合

应用—精确测频

频率整数比—利萨茹图

§ 6.10 相图

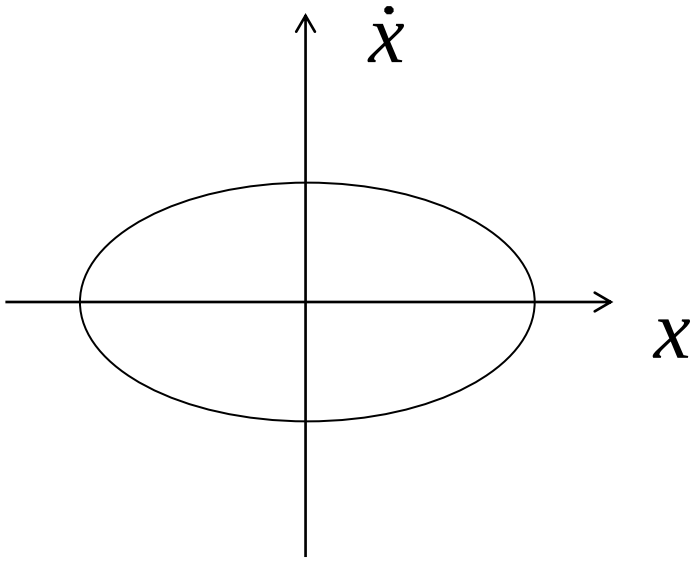


$$\ddot{x} + \omega^2 x = 0 \qquad \omega = \sqrt{\frac{g}{l}}$$

$$\dot{x} = y \qquad x = A\cos(\omega t + \varphi)$$

$$\dot{y} = -\omega^2 x \qquad \dot{x} = -\omega A\sin(\omega t + \varphi)$$

$$x^2 + \frac{y^2}{\omega^2} = A^2$$



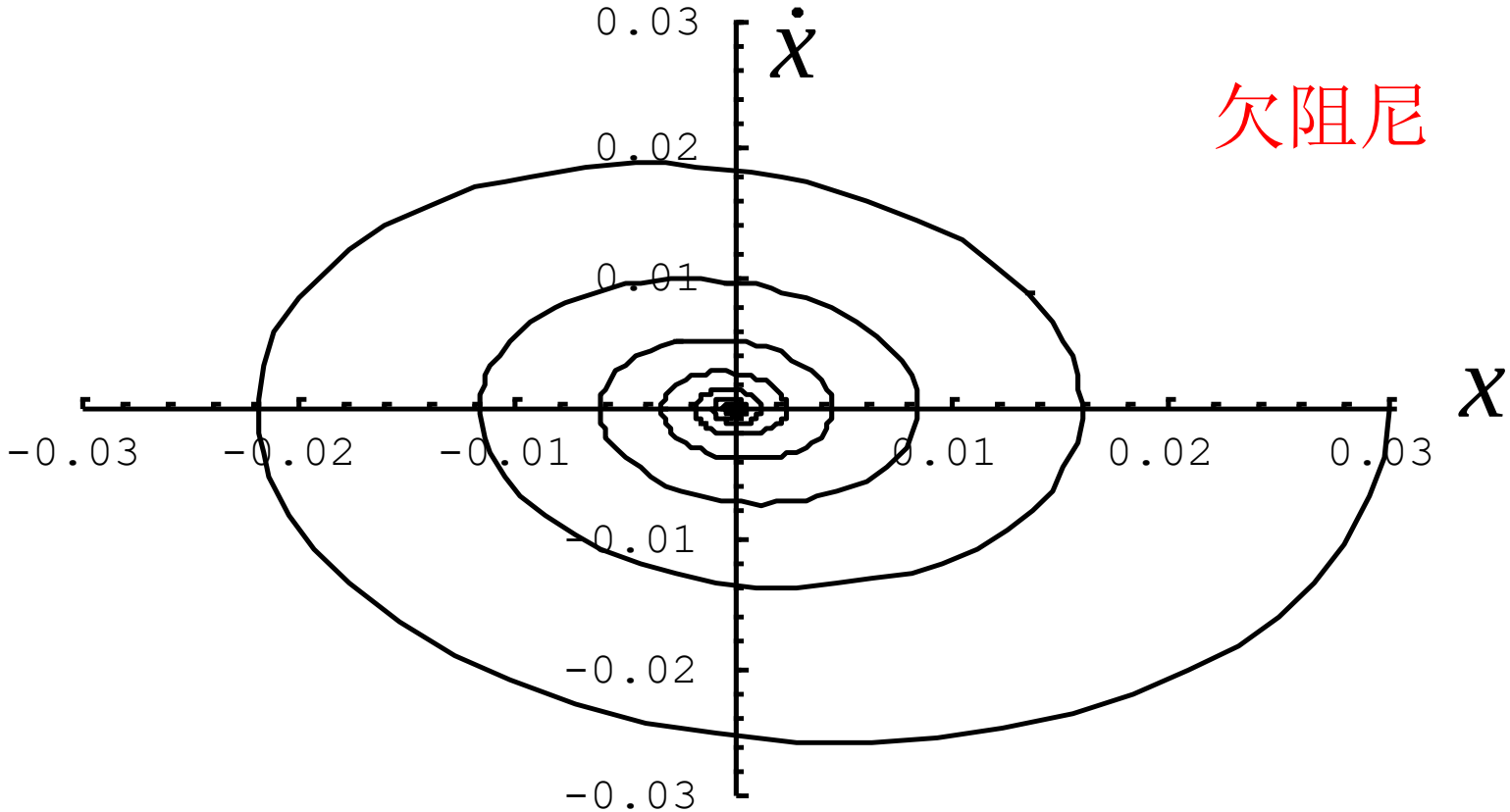
$$\frac{dx}{dy} = -\frac{y}{\omega^2 x} \qquad \omega^2 x^2 + y^2 = c$$

简谐振动相图

阻尼振动

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$$

$$x|_{t=0} = 0.03, \qquad \dot{x}|_{t=0} = 0, \qquad \beta = 0.1, \quad \omega_0 = 1$$



欠阻尼

临界阻尼或过阻尼?

受迫振动

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f\cos\omega t$$

系统最终按ω频率振动

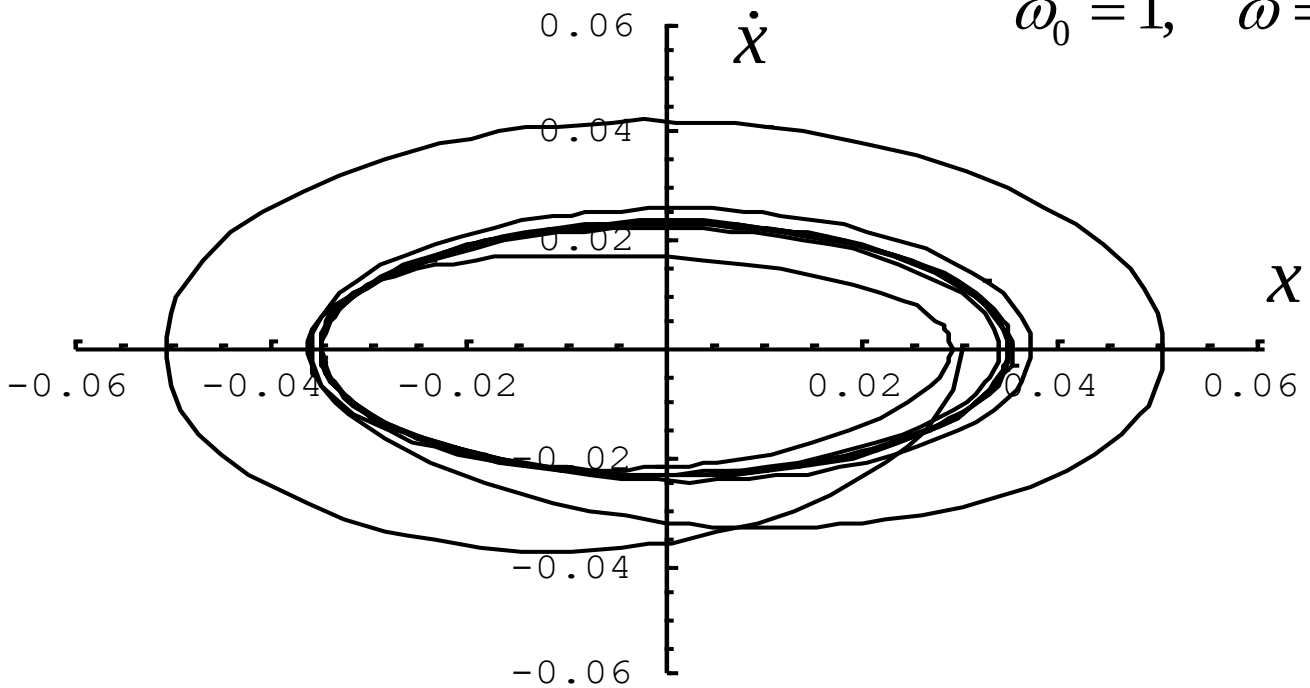
$$x = A\cos(\omega t + \varphi)$$

$$\dot{x} = -\omega A\sin(\omega t + \varphi)$$

$$x|_{t=0} = 0.03, \qquad \dot{x}|_{t=0} = 0,$$

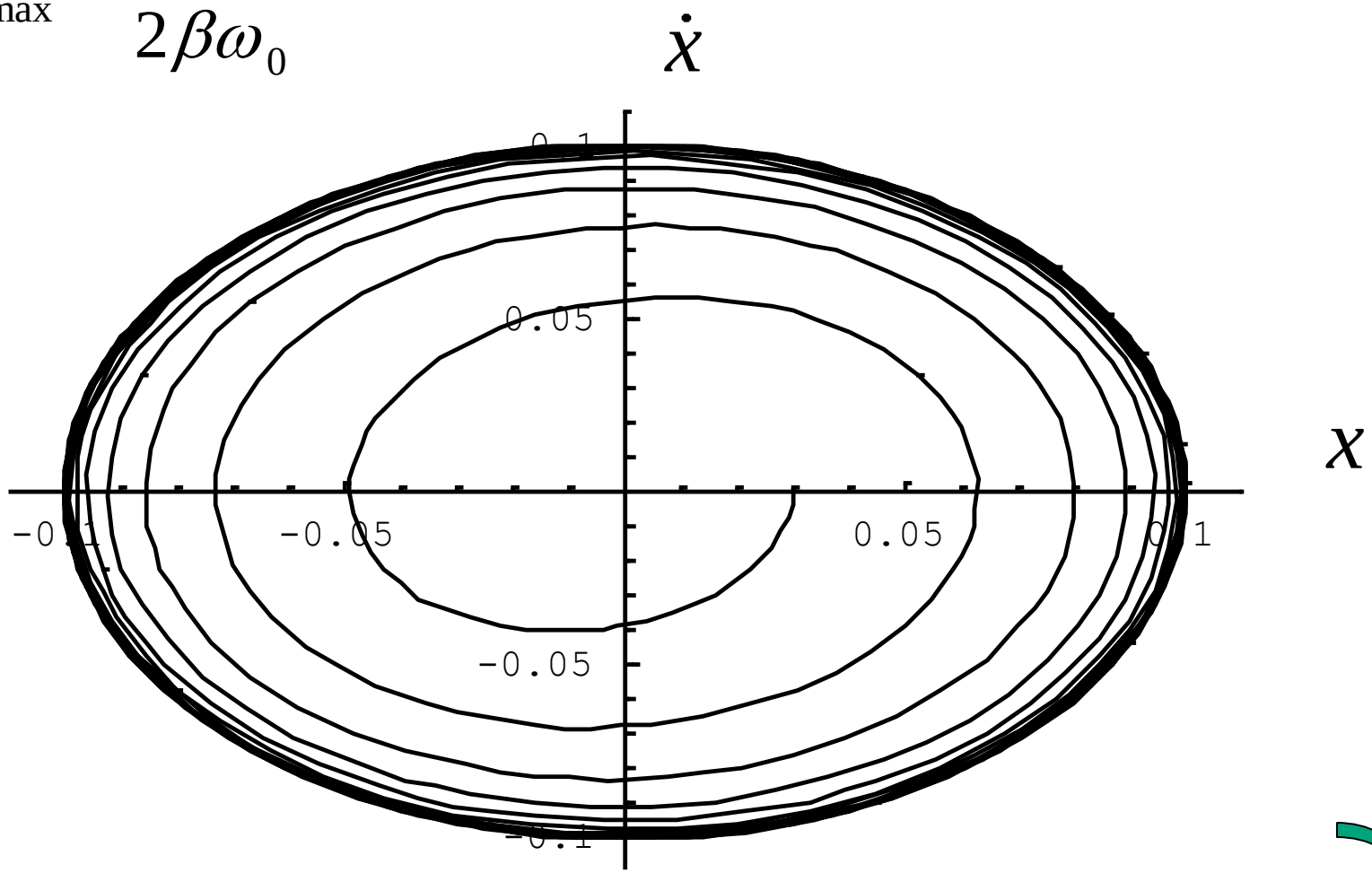
$$\beta = 0.1, \quad f = 0.02,$$

$$\omega_0 = 1, \quad \omega = 0.666,$$



$\omega \approx \omega_0 = 1$ 时产生共振 系统对外界响应最大

$$x_{\max} \approx \frac{f}{2\beta\omega_0}$$



§ 6.11 *非线性振动及混沌

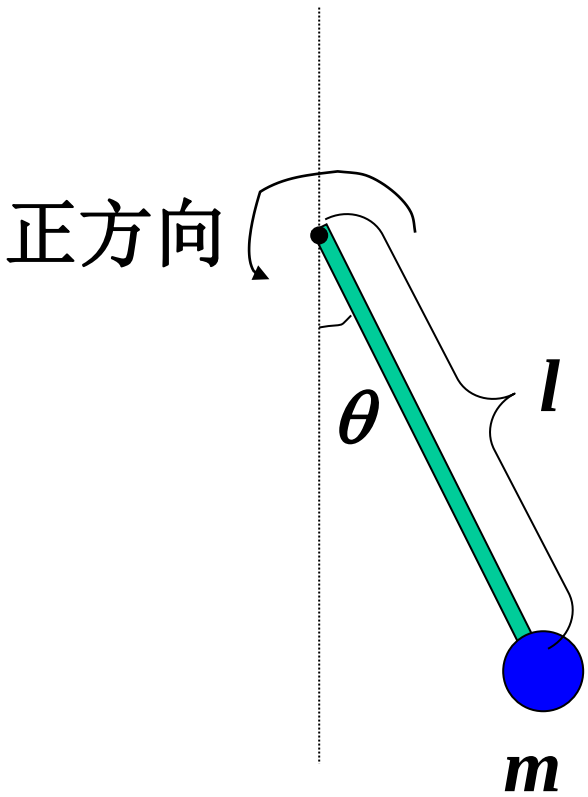
Poincaré 1892 三体问题　发现混沌

E.N.Lorenz 1963 天气模型 数值计算

M.J.Lighthill （噪声）1986:

We collectively wish to apologize for having misled the general public by spreading ideas about the determinism of systems satisfying Newton’s law of motion that, after 1960, were proven to be incorrect……

最简单的非线性振子：轻杆儿一端为轴另一端连一质点



$$\tau_g + \tau_f + \tau_{ext} = I\ddot{\theta}$$

$$\tau_g = -mgl\sin\theta$$

$$\because f_r \propto \upsilon$$

$$\therefore \tau_f = -2\beta\dot{\theta}$$

$$\tau_{ext} = \tau_0 \cos\omega t$$

$$\ddot{\theta} + 2\alpha\dot{\theta} + \omega_0^2 \sin\theta = f\cos\omega t$$

θ很小, 线性, 否则非线性

$$\ddot{\theta} + 2\alpha\dot{\theta} + \omega_0^2 \theta = f\cos\omega t$$

$$\theta = \theta(t)$$

$$\theta|_{t=0} = \xi \qquad \dot{\theta}|_{t=0} = \eta$$

令 $\xi \to \xi + \delta\xi$ 或 $\eta \to \eta + \delta\eta$

经过有限时间 $\theta = \theta(t) + \delta\theta$

$\delta\xi \to 0$时 $\delta\theta \to 0$ 解稳定

$\delta\eta \to 0$

经典力学
可预测性

θ不小

$$\ddot{\theta} + 2\alpha\dot{\theta} + \omega_0^2 \sin\theta = f\cos\omega t$$

$$\theta|_{t=0} = \xi \qquad \dot{\theta}|_{t=0} = \eta$$

$$\theta = \theta(t)$$

令 $\xi \to \xi + \delta\xi$ 或 $\eta \to \eta + \delta\eta$

经过有限时间 $\theta = \theta(t) + \delta\theta$

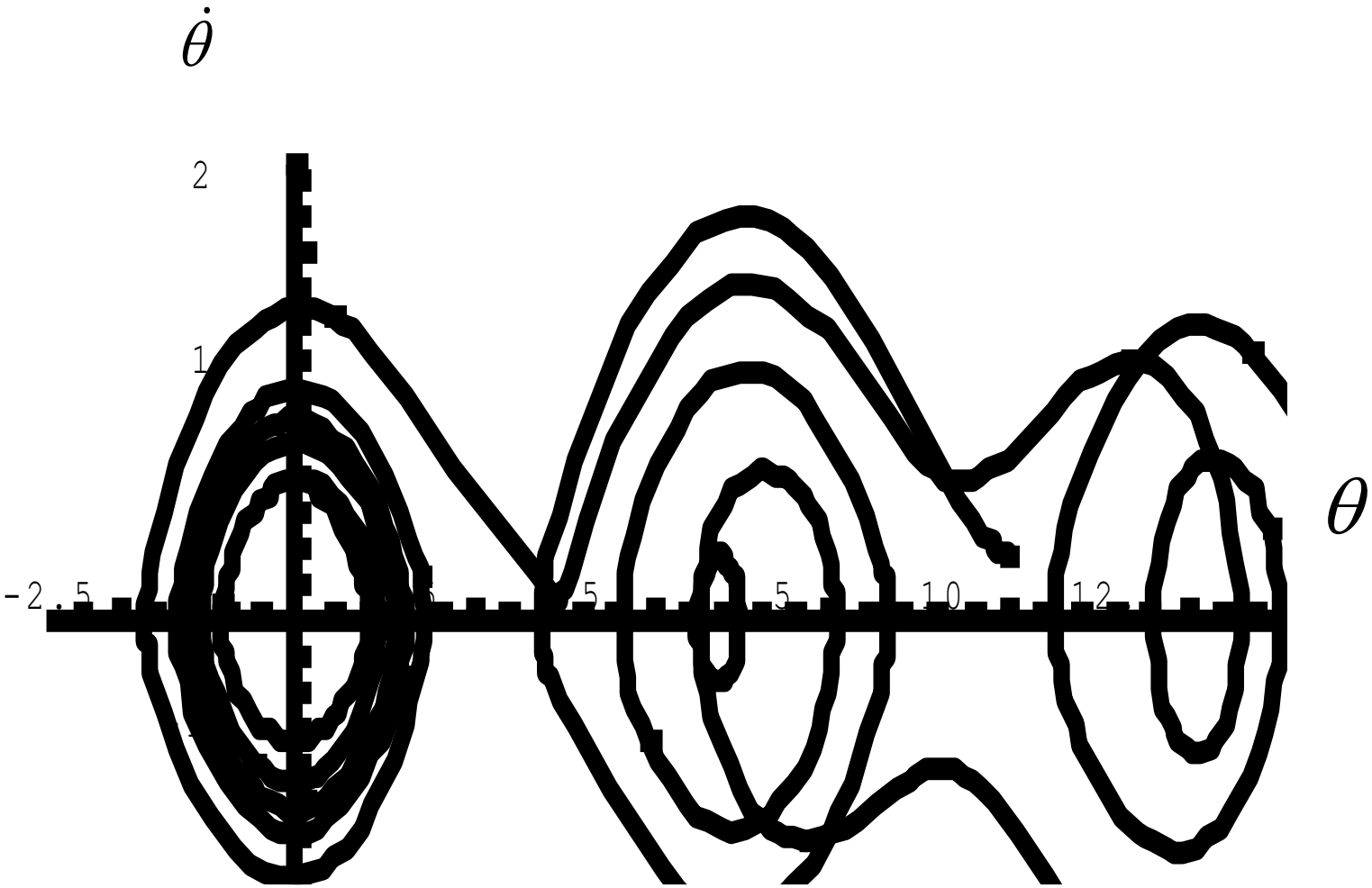
在有些条件下 (f 大)

$\delta\xi \to 0$时 $\delta\theta \not\to 0$ 解不稳定

$\delta\eta \to 0$ 初值敏感，混沌

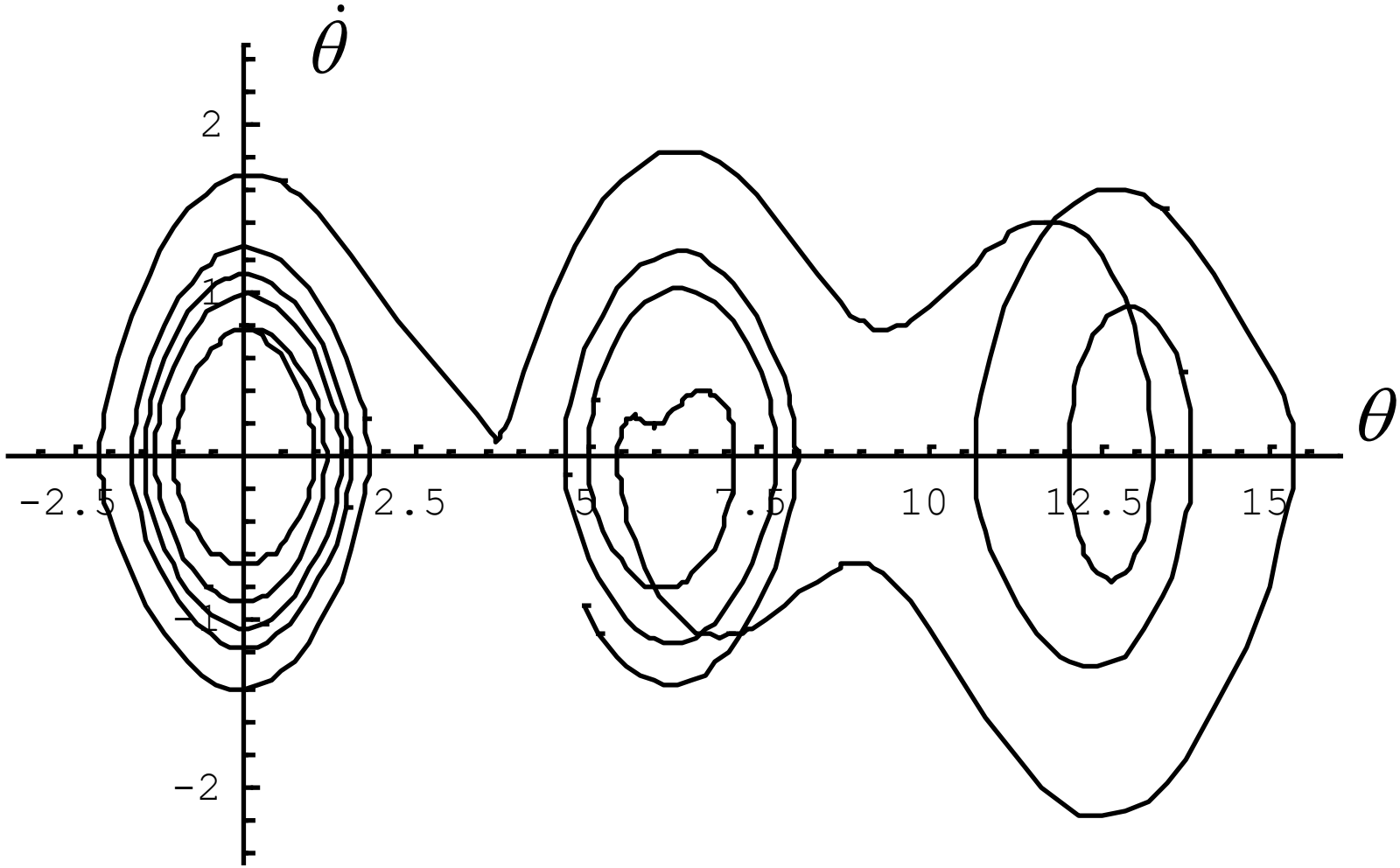
$$\theta|_{t=0} = -0.0885 \qquad \dot{\theta}|_{t=0} = 0.8 \qquad \alpha = 0.1$$

$$f = 0.52 \qquad \omega = 0.666$$



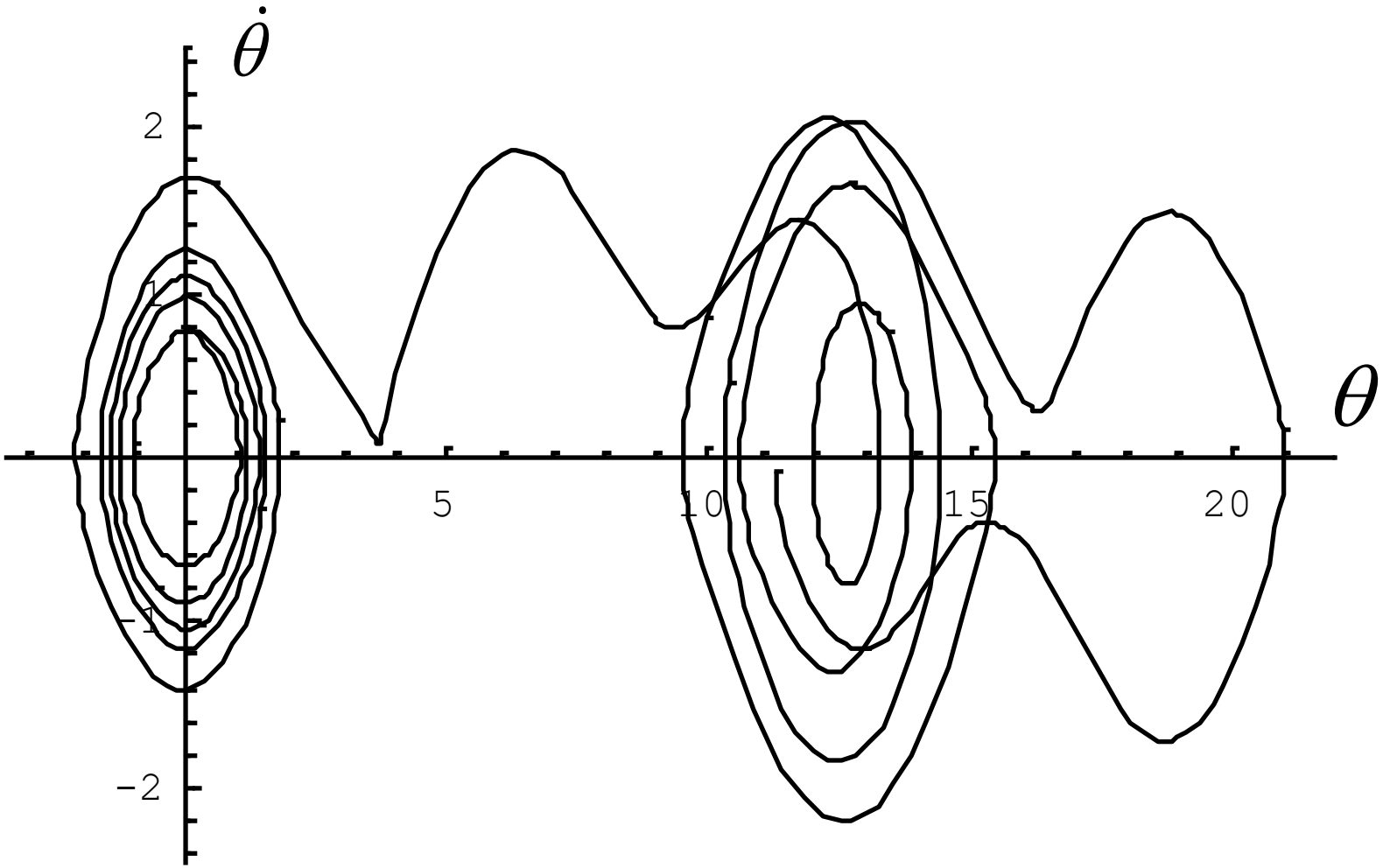
$$\theta\,|_{t=0} = -0.0886 \qquad \dot{\theta}\,|_{t=0} = 0.8 \qquad \alpha = 0.1$$

$$f = 0.52 \qquad \omega = 0.666$$



$$\theta|_{t=0} = -0.0887 \qquad \dot{\theta}|_{t=0} = 0.8 \qquad \alpha = 0.1$$

$$f = 0.52 \qquad \omega = 0.666$$



谢谢！

安 宇